

The True Cost of Food

The price on your shopping receipt is only part of the story — the real cost of cheap food is hidden, vast, and growing.

collateral damage · enervation · externalities · intensive farming · monocultures · pesticides
· fertilisers · mechanisation · hedgerows · faecal · subsidies · arable land · sustainable ·
organic farming · agrochemical · price premium · quantified · feasible · viable

PART ONE · A COST THAT CANNOT BE SEEN AT THE CHECKOUT

A For more than forty years the cost of food has been rising. It has now reached a point where a growing number of people believe that it is far too high, and that bringing it down will be one of the great challenges of the twenty-first century. That cost, however, is not in immediate cash. In the West at least, most food is now far cheaper to buy in relative terms than it was in 1960. The cost is in the collateral damage of the very methods of food production that have made the food cheaper: in the pollution of water, the enervation of soil, the destruction of wildlife, the harm to animal welfare and the threat to human health caused by modern industrial agriculture.

The central paradox of modern food production is this: the methods that have made food cheaper to buy have made it vastly more expensive in every other sense. What the consumer saves at the supermarket checkout, society pays elsewhere — through damaged ecosystems, polluted rivers, declining wildlife, and the enormous public cost of repairing the damage that industrial agriculture leaves behind. To understand how this happened, we need to look at the history of how farming changed — and at the extraordinary scale of the damage that change has caused.

B First mechanisation, then mass use of chemical fertilisers and pesticides, then monocultures, then battery rearing of livestock, and now genetic engineering — the onward march of intensive farming has seemed unstoppable in the last half-century, as the yields of produce have soared. But the damage it has caused has been colossal. In Britain, for example, many of our best-loved farmland birds, such as the skylark, the grey partridge, the lapwing and the corn bunting, have vanished from huge stretches of countryside, as have even more wild flowers and insects. This is a direct result of the way we have produced our food in the last four decades. Thousands of miles of hedgerows, thousands of ponds, have disappeared from the landscape. The faecal filth of salmon farming has driven wild salmon from many of the sea lochs and rivers of Scotland. Natural soil fertility is dropping in many areas because of continuous industrial fertiliser and pesticide use, while the growth of algae is increasing in lakes because of the fertiliser run-off.

■ WHAT INTENSIVE FARMING HAS DESTROYED

- Farmland bird populations — skylark, partridge, lapwing, corn bunting.
- Thousands of miles of **hedgerows** and thousands of ponds.
- Wild salmon driven from rivers and sea lochs by **faecal** pollution.
- Natural soil fertility — undermined by continuous pesticide use.
- Wild flowers and insects across large areas of British countryside.
- Water quality — lakes suffering algae growth from fertiliser run-off.

■ WHAT INTENSIVE FARMING HAS PRODUCED

- Dramatically higher yields of food crops per hectare.
- Lower prices at the supermarket for most staple food products.
- Greater efficiency in labour through **mechanisation** and automation.
- Year-round availability of produce that was once seasonal.
- Enormous agricultural **subsidies** from government and the EU.
- A global food system capable of feeding billions of people.

PART TWO • THE LANGUAGE OF HIDDEN COSTS

C Put it all together and it looks like a battlefield, but consumers rarely make the connection at the dinner table. That is mainly because the costs of all this damage are what economists refer to as **externalities**: they are outside the main transaction, which is for example producing and selling a field of wheat, and are borne directly by neither producers nor consumers. To many, the costs may not even appear to be financial at all, but merely aesthetic — a terrible shame, but nothing to do with money. And anyway they, as consumers of food, certainly aren't paying for it, are they?

"Our food bills are actually threefold. We pay once over the counter, second through our taxes — which fund the enormous subsidies propping up intensive farming — and thirdly to clean up the mess that modern farming leaves behind." — Professor Jules Pretty, University of Essex

PART THREE • PUTTING A FIGURE ON THE DAMAGE

D But the costs to society can actually be **quantified** and, when added up, can amount to staggering sums. A remarkable exercise in doing this has been carried out by one of the world's leading thinkers on the future of agriculture, Professor Jules Pretty, Director of the Centre for Environment and Society at the University of Essex. Professor Pretty and his colleagues calculated the **externalities** of British agriculture for one particular year. They added up the costs of repairing the damage it caused, and came up with a total figure of £2,343m. This is equivalent to £208 for every hectare of **arable land** and permanent pasture, almost as much again as the total government and EU spend on British farming in that year. And according to Professor Pretty, it was a conservative estimate.

E The costs included: £120m for removal of pesticides; £16m for removal of nitrates; £55m for removal of phosphates and soil; £23m for the removal of the bug cryptosporidium from drinking water by water companies; £125m for damage to wildlife habitats, hedgerows and dry stone walls; £1,113m from emissions of gases likely to contribute to climate change; £106m from soil erosion and organic carbon losses; £169m from food poisoning; and £607m from cattle disease. Professor Pretty draws a simple but memorable conclusion from all this: our food bills are actually threefold. We are paying for our supposedly cheaper food in three separate ways: once over the counter, second through our taxes, which provide the enormous subsidies propping up modern intensive farming, and thirdly to clean up the mess that modern farming leaves behind.

■ PROFESSOR PRETTY'S CALCULATIONS — EXTERNAL COSTS OF BRITISH AGRICULTURE	
£1,113m	Greenhouse gas emissions contributing to climate change
£607m	Cattle disease and related agricultural health costs
£169m	Food poisoning and related illness
£125m	Damage to wildlife habitats, hedgerows and dry stone walls
£120m	Removal of pesticides from drinking water
£106m	Soil erosion and organic carbon losses
£55m	Removal of phosphates and soil from water
£23m	Removal of cryptosporidium from drinking water
£16m	Removal of nitrates from drinking water
TOTAL: £2,343m — equivalent to £208 per hectare of arable land	

£2,343m total external cost of British agriculture per year	£208 cost per hectare of arable land and pasture	3× ways in which consumers pay for their food	3 billion: global farming subsidy spend annually
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PART FOUR • IS THERE A BETTER WAY?

F So can the true cost of food be brought down? Breaking away from industrial agriculture as the solution to hunger may be very hard for some countries, but in Britain, where the immediate need to supply food is less urgent, and the costs and the damage of intensive farming have been clearly seen, it may be more feasible. The government needs to create sustainable, competitive and diverse farming and food sectors, which will contribute to a thriving and sustainable rural economy, and advance environmental, economic, health, and animal welfare goals.

G But if industrial agriculture is to be replaced, what is a **viable** alternative? Professor Pretty feels that **organic farming** would be too big a jump in thinking and in practices for many farmers. Furthermore, the **price premium** would put the produce out of reach of many poorer consumers. He is recommending the immediate introduction of a 'Greener Food Standard', which would push the market towards more **sustainable** environmental practices than the current norm, while not requiring the full commitment to organic production. Such a standard would comprise agreed practices for different kinds of farming, covering **agrochemical** use, soil health, land management, water and energy use, food safety and animal health. It could go a long way, he says, to shifting consumers as well as farmers towards a more sustainable system of agriculture.

The debate over the true cost of food is ultimately a debate about who pays, and when. The current system asks consumers to pay less now, while transferring enormous hidden costs to taxpayers, future generations, and the natural world. Professor Pretty's analysis suggests that this bargain is far less attractive than it appears — and that a more **sustainable**, more honest food system would not necessarily cost more overall. It would simply be more transparent about where the costs fall, and more equitable in how they are shared. The Greener Food Standard, if adopted, would represent a first step towards that more honest accounting.

■ TALKING POINTS

The article says consumers 'rarely make the connection at the dinner table'. Why do you think that is?

Professor Pretty says we pay for food three times. Can you explain each payment in your own words?

Do you think a 'Greener Food Standard' is a realistic solution, or does it not go far enough? Why?

Should food that damages the environment cost more at the checkout? Who should decide — governments, markets, or consumers?